

Explainable Multi-Factor Trust Assessment with Calibrated Scores and Factor-Level Abstention: From News Articles to Multimodal Social-Media Content

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1. Abstract

Automated systems for judging online information overwhelmingly return a single verdict — fake or real — produced by a model whose reasoning the reader cannot inspect. A reader handed such a label is left in the same position that misinformation itself creates: asked to believe an assertion they have no means of checking. This project replaces the verdict with an assessment. Every item of content is examined along six independent axes — the publishing outlet’s record, the honesty of its headline, the fairness of its language, whether its claims can be traced to a nameable source, the persuasion techniques it employs, and whether independent outlets report the same facts — and each axis returns not only a score but the evidence that produced it: sentences quoted verbatim from the text, the journalistic checks the item passes and fails, and links to the corroborating coverage that was actually retrieved. The axis scores combine through a published weighted formula rather than a second opaque model, so the arithmetic behind a trust score is auditable by the reader.

Two properties separate this work from prior explainable detectors. First, assessment abstains at the level of the individual check: an axis that finds nothing to weigh contributes zero, states in plain language why it could not contribute, and releases its weight proportionally to the axes that did produce signal, so a check that failed to run is never presented as a check that ran and found nothing wrong. Second, the resulting score is calibrated: a Platt-scaling step reduced expected calibration error from 0.277 to 0.073 on a leakage-controlled ISOT evaluation, so a stated confidence can be read as a frequency. In that evaluation an earlier four-axis configuration reached 94.7% accuracy and 0.987 ROC-AUC against 82.0% and 0.838 for a single-prompt baseline over the same underlying language model, indicating that the gain arises from the decomposition rather than from model scale. The system is deployed as a working web application that analyses live news and tolerates provider congestion, model withdrawal, and total model unavailability without fabricating confidence.

The proposed extension generalises the same framework beyond news articles to multimodal social-media artefacts — an image-and-caption post, a spoken claim in a video — where the normal condition is that some evidence is simply absent. The abstention mechanism, already central to the present system, is precisely what allows a trust score over such content to remain interpretable when a modality, a transcript, or corroborating coverage does not exist.

Keywords — *explainable artificial intelligence; misinformation assessment; evidence-grounded verification; confidence calibration; selective abstention; propaganda technique detection; multimodal content credibility*

2. Base Paper

S. Amri, H.-C. Mputu Boleilanga, and E. Aïmeur, “*ExFake: Towards an Explainable Fake News Detection Based on Content and Social Context Information*,” **arXiv:2311.10784**, **November 2023**.

What the base paper does. ExFake judges the veracity of an online post by fusing three families of signal — the content of the post itself, social-context information (the credibility and historical behaviour of the users circulating it), and evidence drawn from trusted external entities such as fact-checking websites and named-entity records. Rather than emitting a bare label, it couples the decision to an explainable-AI assistant intended to help an ordinary social-network user reason about doubtful information. It is evaluated on a real-world dataset and reported to outperform its baselines.

Why this paper, and not another. ExFake is the closest published match to this project on the dimensions that define it.

Dimension	ExFake (base paper)	This project
Objective	Help a non-expert reader judge a post, not merely classify it	Same
Method	Fusion of several heterogeneous signal families rather than one classifier	Same
External grounding	Evidence from outside the post itself	Same — live retrieval from independent outlets
Audience	End user, via an explanation assistant	End user, via an inspectable trust report

Alternatives considered and rejected as the base. Three recent systems were weighed against ExFake.

- **VeraCT Scan [3]** (2024) performs an internet-wide search for corroborating and conflicting reports and justifies its verdict — technically the nearest neighbour to this project’s verification factor. It was not chosen because it already occupies that ground, leaving a narrower delta: the contributions that remain — factor decomposition with exposed weights, abstention, calibration, specified degradation — are better presented as improvements on a system whose scope is the full reader-facing assessment.
- **TELLER [2]** (Findings of ACL 2024) is explainable through logic rules rather than through retrieved evidence, so its explanation is a different object from the one this project produces.
- **TRUST-VL [6]** (EMNLP 2025) is the strongest recent multimodal explainable system, but it is a vision-language model trained on a 198K-sample instruction corpus; adopting it as the base would frame this project as an under-resourced attempt at the same thing, whereas the contribution here is architectural rather than a matter of training scale. It is therefore treated as the state of the art against which the proposed multimodal extension is positioned (Section 5), not as the paper being extended.

Currency of the literature. Every comparative reference in Section 7 is from 2023 or later; the two pre-2020 entries are cited only as the provenance of standard techniques — expected calibration error and Platt scaling — and are listed separately as such.

3. Limitations of the Base Paper

#	Limitation in ExFake	Consequence
L1	Depends on social-context signals — user credibility and circulation history	No assessment is possible on first sight of an item, which is exactly when a reader needs one
L2	Signals are fused into a decision; the contribution of each is not exposed as an auditable quantity	The reader sees an explanation but cannot audit how much each signal moved the result
L3	Behaviour when a signal family yields nothing is unspecified	A check that could not run is indistinguishable, at the interface, from a check that ran and found nothing wrong
L4	No calibration of the confidence attached to a verdict	A stated confidence cannot be interpreted as a probability
L5	Explanation is at the level of the decision, not of the specific text that caused it	The reader cannot verify the explanation against the item’s own words
L6	Evaluated offline; no account of degradation when an external dependency is unavailable	Failure behaviour in deployment is undefined

4. Proposed System and Points of Novelty

Novelty is stated at a level a reviewer can verify. Claims N1–N5 are implemented and measured in the present system; N6 is the proposed extension.

N1 · Factor-level abstention with weight redistribution. Let F be the set of factors with prior weights $w(f)$ summing to one, and let I be the subset that produced informative evidence. Factors outside I abstain; the surviving weights are renormalised as $w'(f) = w(f) / \sum w(g)$ over g in I , and the trust score is $S = \sum w'(f) \cdot s(f)$ over f in I . An abstaining factor is rendered to the reader with no score, a hatched bar, and a written reason.

Distinction from prior art. Selective prediction in fact-checking abstains at the level of the whole verdict — a system returns “uncertain” in place of true or false [9]. Abstention inside a composite score, in which one evidence family withdraws, surrenders its weight to the others, and explains its own absence to the reader while the assessment still completes, has not to our knowledge been described. The defect this prevents was observed directly: four content factors that answered a failed model call with a neutral 5/10 produced a confident “Mostly fine” report assembled entirely from placeholders.

N2 · A calibrated, reader-facing trust score. Expected calibration error is reported before and after Platt scaling ($0.277 \rightarrow 0.073$), fitted on a validation half and reported on a held-out half. Calibration is routine in the machine-learning literature, but it is not established practice among the language-model fact-checking systems surveyed in [5], and it is absent from the base paper — yet it is what allows a trust score to be read as a frequency rather than as a sentiment.

N3 · Decomposition as the source of accuracy. Holding the language model, the article sample, and the prompt budget fixed, routing the judgment through specialised factors instead of one holistic prompt improved accuracy by 12.7 points and ROC-AUC by 0.149. The claim under test concerns architecture rather than model capability, and the ablation ordering is monotonic.

N4 · Two reader-facing factors not previously deployed as such. Source transparency scores an article against a five-point journalism checklist and quotes back the unattributable appeals it leans on (“experts say”). Persuasion technique applies the taxonomy standardised by the SemEval shared tasks — most recently the twenty-three techniques of SemEval-2023 Task 3 [8] — which has to date existed

as a research benchmark, as a factor that names the technique to a non-specialist reader and quotes the span exhibiting it. Every factor must quote the text it acted on, so each explanation is falsifiable against the article.

N5 · Specified degradation. The language model is reached through a gateway that distinguishes model withdrawal, provider congestion, and provider exhaustion, and every content factor has a deterministic lexicon-and-pattern fallback. When no model is reachable the report declares itself degraded at both summary and per-factor level. With zero model calls the system still scored a fabricated article 2.3/10 and ordinary attributed reporting 9.8/10.

N6 · Provenance- and modality-aware extension to social-media content (*proposed* — Section 5).

5. Proposed Extension: Trust Assessment over Multimodal Social Content

Motivation. The reader who most needs an assessment does not usually encounter a claim as a news article. They encounter it as a screenshot-style Instagram post, a caption over an image, or a sentence spoken in a video. The present pipeline cannot be pointed at those artefacts, and the reviewers' observation that the work should reach further is accepted.

What is proposed. A normalisation layer that converts a social-media artefact into the canonical object the six factors already consume — text, a title-like claim, and a provenance record — by extracting caption text, reading text rendered inside an image, and transcribing spoken audio. Two factors then generalise rather than being replaced:

- **Source reputation** becomes *account provenance*: an outlet's factual-reporting record gives way to the observable properties of a posting account — age, verification state, stated affiliation, and whether it is a recognised outlet's own channel.
- **Cross-source verification** becomes *claim-to-coverage matching for claims that never appeared in text form*: a statement spoken in a video is extracted, written down, and checked against independent reporting exactly as an article's claim is.

Where the novelty lies, stated precisely. Multimodal misinformation detection is an active field, and a claim of novelty resting on the use of images or video would not survive scrutiny: LVLM-based systems including TRUST-VL [6] and LEMMA [7] already classify multimodal items and emit rationales. The distinguishing property proposed here is **modality-aware abstention**. Those systems assume a complete modality bundle and produce a label with a rationale; none defines what the output means when a modality is missing — and on social platforms missing modalities are the normal case, not the exception. A post may carry no caption, a video no usable transcript, an account no recoverable provenance, a claim no independent coverage. A system that silently degrades under those conditions returns a trust score whose meaning changes without notice.

The abstention mechanism of N1 answers exactly this. A modality that cannot be read, or a factor that has nothing to weigh, withdraws from the score, releases its weight, and tells the reader what could not be checked. The resulting claim — an explainable, calibrated trust assessment that remains interpretable under partial observation of a multimodal artefact — is, to our knowledge, unclaimed, and it follows from a property this project already implements rather than being bolted on.

Deliberately out of scope. Synthetic-media forensics — deepfake and generated-image detection — is not proposed. It is a distinct research problem with its own failure modes, and claiming it would weaken the submission. The architecture admits such a detector as an additional factor with a weight and an abstention rule whenever a reliable one becomes available, which is the correct way for this system to absorb it.

6. Evaluation Plan

Claim	How it is evidenced
Decomposition outperforms a single prompt	Leakage-controlled ISOT benchmark; stratified seeded sample; identical article set per configuration; single-prompt baseline over the same model
Every factor contributes	Leave-one-out ablation across all six factors
Confidence is honest	Ten-bin expected calibration error, raw and Platt-scaled, fitted on validation and reported on a held-out half
Explanations are faithful	Verbatim-span audit — every quoted span must occur in the source text
Failure behaviour is safe	Language model made unreachable; discriminative separation of the degraded pipeline measured
Verification is grounded	Live-news study with manual annotation of verdict correctness, reporting coverage rate and precision

Leakage control is explicit. The genuine articles of the ISOT corpus [10] originate from a single news agency, so outlet identity alone predicts the label; source identity is therefore hidden from the pipeline during evaluation and agency prefixes are stripped from the text, and the reported figures are content-derived.

Status note. The 94.7% / 82.0% comparison and the $0.277 \rightarrow 0.073$ calibration figures were measured on a four-factor configuration using a language model the provider has since withdrawn. They are reported as the measurement that motivated the architecture; a re-measurement of the present six-factor configuration on the current model is the outstanding task before submission.

7. References

Primary and comparative work (2023–2026)

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Dataset

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Methodological foundations — cited for the provenance of standard techniques rather than as contemporary comparison

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Appendix A — Short Abstract (150 words)

Automated misinformation systems return a label from an opaque model, which gives a reader nothing they can check. This project assesses content instead of classifying it: six independent factors — outlet record, headline honesty, language fairness, source transparency, persuasion technique, and cross-source verification — each return a score together with the evidence behind it, quoted from the text or linked to retrieved coverage, and combine through a published weighted formula. Two properties are central. A factor with nothing to weigh abstains, surrenders its weight to the factors that produced signal, and tells the reader why, so an unrun check is never mistaken for a passed one. The score is calibrated, with expected calibration error reduced from 0.277 to 0.073. On a leakage-controlled ISOT evaluation the decomposed pipeline reached 94.7% accuracy against 82.0% for a single-prompt baseline over the same model. The proposed extension carries the same abstention guarantee to multimodal social-media content, where missing evidence is the normal condition.